

The empirical recurrence for OEIS A259892, recovered from its tail

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Abstract

OEIS A259892 records “Empirical recurrence of order 84” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 116$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 86$. Its last coefficient is $c_{84} = 14 \neq 0$, so no further tail satisfies anything shorter; any order-84 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A259892 is “Number of $(n+2) \times (6+2)$ 0..1 arrays with each 3×3 subblock having clockwise perimeter pattern 00000000 00000001 or 00000011.”. It has offset 1 and begins

2212, 13704, 151483, 954624, 5055255, 36899517, 265544204, 1657505754, ...

The entry states:

Empirical recurrence of order 84 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 11:55:50, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 116$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 116$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 84: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 84.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 232$ exact terms from $n = 2$ on, it returns order 84 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{84} c_i a(n-i),$$

c_1	1	c_{22}	−18199921493	c_{43}	3490835039758	c_{64}	2882671898
c_2	8	c_{23}	−27865686493	c_{44}	5282827842374	c_{65}	2102071737
c_3	145	c_{24}	−29047294225	c_{45}	5209504211290	c_{66}	5909793334
c_4	679	c_{25}	19481206089	c_{46}	54010866038	c_{67}	−518419978
c_5	−123	c_{26}	130115935718	c_{47}	−2544022963729	c_{68}	−119443391
c_6	−6761	c_{27}	214262724718	c_{48}	−3239673853141	c_{69}	−233348444
c_7	−40760	c_{28}	254526748562	c_{49}	−3214591065072	c_{70}	−350674578
c_8	−114515	c_{29}	39278124481	c_{50}	321878025723	c_{71}	115608925
c_9	121117	c_{30}	−503208104391	c_{51}	1451111153021	c_{72}	−37352745
c_{10}	1207092	c_{31}	−954970609548	c_{52}	1210358987051	c_{73}	19951925
c_{11}	3369003	c_{32}	−1244361938386	c_{53}	1127840677999	c_{74}	7281755
c_{12}	5127758	c_{33}	−706637403176	c_{54}	−426864926897	c_{75}	−4842849
c_{13}	−11841883	c_{34}	992747110127	c_{55}	−559264912405	c_{76}	3079407
c_{14}	−62395811	c_{35}	2448816606406	c_{56}	−224750243375	c_{77}	−1403301
c_{15}	−114631186	c_{36}	3431979160190	c_{57}	−186039326575	c_{78}	354649
c_{16}	−112380945	c_{37}	2729340690455	c_{58}	210200392822	c_{79}	−101350
c_{17}	344439648	c_{38}	−821113447118	c_{59}	117523857115	c_{80}	−117
c_{18}	1439513387	c_{39}	−3634228422419	c_{60}	3816731194	c_{81}	4686
c_{19}	2232661059	c_{40}	−5450732944717	c_{61}	3612287223	c_{82}	−1877
c_{20}	2048290310	c_{41}	−5022008654434	c_{62}	−48170928950	c_{83}	668
c_{21}	−4203434375	c_{42}	100139959268	c_{63}	−8755530028	c_{84}	14

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 86$, and it is the recurrence OEIS A259892 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{84} - c_1x^{84-1} - \dots - c_{84}$. Its constant term is $-c_{84} = -14$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 84 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 84, so $\deg q = 84$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 84, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 17 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 84, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 14.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-84 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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